Result 1 (exact maps vanish on the image). [Mathlib exact pairs, Function.Exact.apply_apply_eq_zero]
Let K be a type. Let *field* be a field structure on K. Let M, N and P be elements of Mod_K. Let f be a linear map from M to N. Let g be a linear map from N to P. Let x be an element of M. Assume $\mathrm{exact}(f, g)$ (`exactness of f and g`). Then

$$g(f(x)) = 0.$$

Result 2 (exact sequence interior at each position). [Topology foundations, exact sequence definition]
Let K be a type. Let *field* be a field structure on K. Let S be a sequence of modules over K. Let n be a natural number. Assume interior exactness of S (`exactness`). Then

$$\mathrm{exact}_{n+1}(S).$$

Result 3 (exactness before a zero map). [Mathlib exact pairs, LinearMap.exact_zero_iff_surjective]
Let K be a type. Let *field* be a field structure on K. Let M, N and P be elements of Mod_K. Let f be a linear map from M to N. Then

$$\mathrm{exact}(f, 0) \text{ if and only if } f \text{ is onto.}$$

Result 4 (terminal exactness includes the interior). [Topology foundations, terminal exactness definition]
Let K be a type. Let *field* be a field structure on K. Let S be a sequence of modules over K. Assume S is exact and ends in zero (`exactness`). Then

$$\text{interior exactness of } S.$$

Result 5 (terminal exactness covers degree zero). [Topology foundations, terminal exactness definition]
Let K be a type. Let *field* be a field structure on K. Let S be a sequence of modules over K. Assume S is exact and ends in zero (`exactness`). Then

$$S.\mathrm{arrow}(0) \text{ is onto.}$$

LLM-assisted definitions and supporting proofs (gpt-6, 2026-11-10). Interior exactness concerns positive degrees; terminal exactness separately requires surjectivity onto degree zero. No Mayer–Vietoris theorem is asserted.